ELSEVIER

Physics of the Earth and Planetary Interiors 106 (1998) 93–101

PHYSICS OF THE EARTH AND PLANETARY INTERIORS

Geomagnetic paleosecular variation in Easter Island, the southeast Pacific

Masako Miki [a,*], Hiroo Inokuchi [a,1], Satoru Yamaguchi [a], Jun-ichi Matsuda [b], Keisuke Nagao [c], Nobuhiro Isezaki [d], Katsumi Yaskawa [a,2]

[a] *Department of Earth and Planetary Sciences, Faculty of Science, Kobe University, Kobe 657, Japan*
[b] *Department of Earth and Space Sciences, Faculty of Science, Osaka University, Osaka 560, Japan*
[c] *Laboratory for Earthquake Chemistry, Faculty of Science, University of Tokyo, Tokyo 113-0033, Japan*
[d] *Department of Earth Sciences, Faculty of Science, Chiba University, Chiba 260, Japan*

Received 20 July 1996; received in revised form 26 March 1997; accepted 2 October 1997

Abstract

We collected more than 250 samples from 45 sites in lava flows of the three volcanoes on Easter Island. The K–Ar whole rock dating was attempted on 13 sites. All of the ages obtained are younger than 0.8 Ma. Reliable paleomagnetic directions were obtained from 34 sites. The mean VGP position is 89.2°N and 239.1°E with a 95% confidence value of 3.8°. The angular dispersion value of Easter Island VGPs was calculated to be 11.8° with an upper limit of 14.1° and a lower limit of 9.8°. The angular dispersion value is significantly lower than the values predicted by the theoretical secular variation models. The pacific dipole window appears to exist also in the southeast Pacific. © 1998 Published by Elsevier Science B.V.

Keywords: Geomagnetic paleosecular variation; Easter Island; K–Ar rock dating

1. Introduction

The geomagnetic secular variation for the last few million years in the Pacific region is characterized by an anomalously low value for the angular dispersion. This low value has been observed in paleomagnetic data from Hawaii (e.g., Doell and Cox, 1961, 1963, 1965, 1971, 1972; McWilliams et al., 1982; Peng and King, 1992), Southeast Australia (Aziz-ur-Rahman, 1971), and the Pacific islands of Pagan (U.S.–Japan Paleomagnetic Cooperation Program in Micronesia, 1975) and Norfolk (Aziz-ur-Rahman and McDougall, 1973). This phenomenon is generally ascribed to the presence of the Pacific dipole window (e.g., Doell and Cox, 1971).

Although its presence is generally acknowledged in Hawaii, the distribution of the Pacific dipole window, in particular to the southeast is not well known despite numerous studies. The purpose of the present study is to estimate the paleosecular variation on Easter Island and to constrain the southeastward distribution of the Pacific dipole window. Studies on the Galapagos Islands (Doell and Cox, 1972), Soci-

* Corresponding author.
[1] Present address: Research Center for Inland Seas, Kobe University, Awaji-cho, Tsuna-gun 656-24, Japan.
[2] Present address: Osaka Junior College, Osaka 587, Japan.

0031-9201/98/$19.00 © 1998 Published by Elsevier Science B.V. All rights reserved.
PII S0031-9201(97)00106-4

ety Islands (Duncan, 1975) and Hiva-Oa Island (Katao et al., 1988) have shown that the secular variation is concordant with that expected from the standard models (Cox, 1962, 1970; McFadden and McElhinny, 1984; McFadden et al., 1988). In contrast, a recent study (Shibuya et al., 1995) reports anomalously low secular variation over a wide area including New Zealand, the Marquesas and Society Islands. Easter Island (27.1°S, 109.2°W) is in the southeast margin of the disputed area of low angular dispersion. Data from this area is important for defining the size and extent of the Pacific dipole window, which can in turn give information to aid our understanding of the Earth's interior and the origin of the Earth's magnetic field.

A previous study on Easter Island (Isaacson and Heinrichs, 1976) indicates slightly low values of paleosecular variation. However, the rock samples were collected from only one of three independent volcanoes on Easter Island and the existence of the Pacific dipole window at the location is still unclear. In an attempt to resolve this question, we collected samples from all three volcanoes and here we report the results of the subsequent geochronological and paleomagnetical analyses.

2. Geology and sampling

Easter island is an isolated volcanic island, roughly triangular in shape and with an area of 160 km^2 (Fig. 1). It is located about 500 km east of the East Pacific Rise and rises from the Sala y Gomez Ridge. Details of the geology and petrology of the island are described by Baker (1966), Baker et al. (1974) and Clark and Dymond (1977). The island consists of three volcanoes: Poike Volcano that forms the easternmost part of the island and Rano Kau Volcano in the southwest are relatively old in comparison with the products of Mt. Terevaka and its satellites. Some geochronological data between 0.2 and 3 Ma have been reported from volcanic rocks of the island (Baker et al., 1974; Clark and Dymond, 1974, 1977; Gonzalez-Ferran et al., 1974; Kaneoka and Katsui, 1985). Terevaka Volcano is a complex fissure vol-

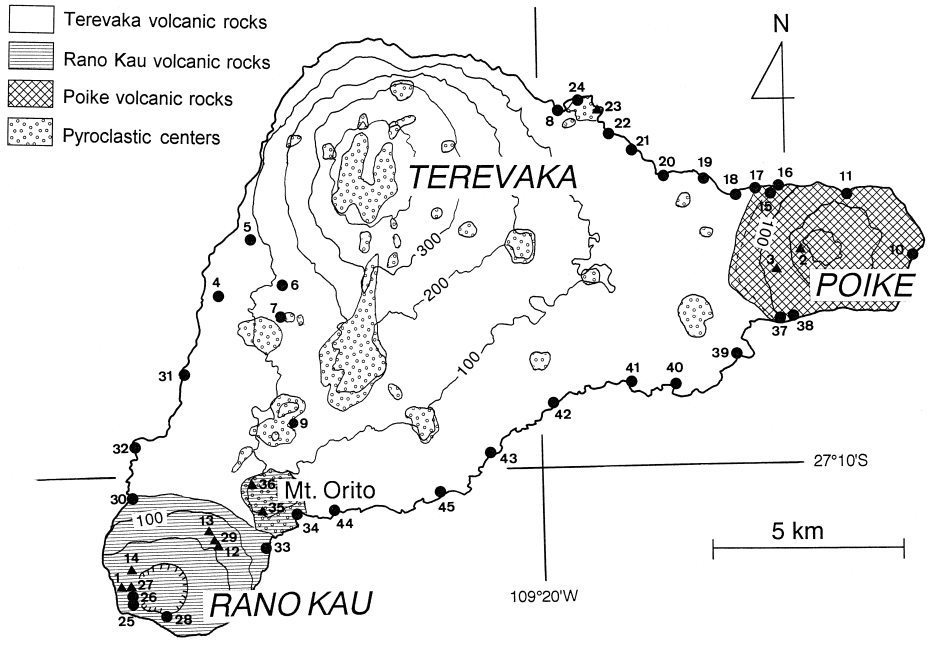


Fig. 1. Schematic geological map of Easter Island with sampling localities. Circles (triangles) with numbers indicate the sites for paleomagnetic (geochronological) study.

cano and has at least seventy subsidiary centers making it difficult to correlate individual flows across the island. The lavas show a wide compositional spread from tholeiites and olivine tholeiites to hawaiites, mugearites, benmoreites, trachytes and rhyolites.

We collected more than 250 samples from 45 sites for geochronological and paleomagnetic measurements. There are nine sites in the flows of Poike Volcano, 14 sites in Rano Kau Volcano and 22 sites in Terevaka Volcano. Five independently oriented hand samples were taken from each site using a magnetic compass to orient them. There was no influence of rock magnetization on the magnetic compass that could be recognized by using a transit: there was no discernible difference in compass directions between measurements close to and far from the rocks.

Although the exact stratigraphic order of the flows is not known due to their complex geometries, field observations suggest the sampling sites are randomly distributed on the lava sequences: each independent flow includes only one site and sampling sites are scattered around the eruption centers ensuring that the flows are randomly sampled at least on the surface.

3. Geochronology

K–Ar whole rock dating was attempted on volcanic rock samples from 13 sites; three sites of Poike Volcano, seven sites of Rano Kau Volcano and three sites from Terevaka Volcano. Prior to the dating, thin sections were examined. The samples are fresh enough to determine the age of the eruption according to the criteria discussed by Mankinen and Darlymple (1972).

Whole rock samples were prepared for argon analysis by crushing to 60–100 mesh size. A portion of each sample was further ground in an agate mortar and used for potassium analysis. Argon analyses were carried out with the mass spectrometer installed in Okayama University of Science. Argon was extracted by fusing 0.5–0.9 g at about 1600°C in a high vacuum line. Prior to fusion, all samples for argon analysis were baked out at 150°C for a whole day in vacuum to reduce the atmospheric Ar component on the surface. The amount of radiogenic ^{40}Ar was determined by the isotopic dilution method using the ^{38}Ar tracer for Poike and Terevaka samples. However, the amounts of radiogenic ^{40}Ar were too small to be determined precisely for these samples except for E16. For the Rano Kau samples, we measured Ar concentrations and isotopic ratios by comparing the peak height of ^{36}Ar for the sample with that of a pipetted amount of air but without using the ^{38}Ar tracer. Potassium content was determined at Kobe University by duplicated atomic absorption spectrometry using the addition of a strontium solution (Terashima, 1970) with the JB-1 standard sample ($K_2O = 1.42\%$) as the standard solution. The errors in K content were 1 to 3% for the duplicated analyses. Ages and their errors were calculated using the method of Nagao and Itaya (1988) with the decay constants of Steiger and Jäger (1977).

The results of the K–Ar dating are listed in Table 1. Absolute ages ranging between 0.1 Ma and 0.7 Ma were obtained from six sites of Rano Kau and one site of Poike Volcanoes. For the other sites, the

Table 1
Results of K–Ar dating on lava flows from Easter Island

Site number	K (wt.%)	$^{40}Ar/^{36}Ar$	^{40}Ar-rad. (10^{-8} cm^3 g^{-1})	Age (Ma)
Poike Volcano				
E10[a]	0.409		< 0.31	< 0.20
E16	0.578	(301 ± 1)	0.31 ± 0.23	0.14 ± 0.10
E37[a]	0.432		< 0.58	< 0.34
Rano Kau Volcano				
E1	3.48	422 ± 3	4.58 ± 0.47	0.34 ± 0.03
E12	3.33	357 ± 1	5.96 ± 0.61	0.46 ± 0.05
E13	3.36	344 ± 2	8.82 ± 0.94	0.68 ± 0.07
E27	3.55	665 ± 17	4.77 ± 0.52	0.35 ± 0.04
E35	3.38	408 ± 6	4.32 ± 0.49	0.33 ± 0.04
E36	3.34	309 ± 4	1.48 ± 0.46	0.11 ± 0.04
Terevaka Volcano				
E20[a]	0.617		< 0.37	< 0.15
E22[a]	1.07		< 3.2	< 0.77
E30[a]	0.488		< 0.32	< 0.17
E43[a]	1.00		< 0.47	< 0.12

[a] The $^{40}Ar/^{36}Ar$ ratios were atmospheric for these samples. Hence, we gave the upper limit of ages by assuming that 2% of the total ^{40}Ar were radiogenic for these samples. The constants used in the calculation are $\lambda_\beta = 4.962 \times 10^{-10}$ yr^{-1}, $\lambda_\varepsilon = 0.581 \times 10^{-10}$ yr^{-1}, and $^{40}K = 1.167 \times 10^{-4}$ atom per atom of natural potassium (Steiger and Jäger, 1977).

amounts of radiogenic ^{40}Ar in total ^{40}Ar were too small to determine the absolute age. This is due to the low K contents in these samples. Therefore, we only estimated the upper limits of the ages for these samples by assuming that at most 2% of the total ^{40}Ar was radiogenic. This is because the mass spectrometer in Okayama University of Science had a detector limit which would clearly detect a 2% difference in Ar isotopic ratios from the atmospheric value.

The obtained ages are all younger than 0.8 Ma, including the upper limits estimated for the Terevaka and Poike Volcanoes. These young ages are concordant with most of the previous studies; Kaneoka and Katsui (1985) from Rano Kau and Poike, Gonzalez-Ferran et al. (1974) from Rano Kau and Terevaka, and Clark and Dymond (1974), Clark and Dymond (1977) from Poike Volcanoes. No difference in age was recognized between the three volcanoes, with all showing continuous activity during the last 0.8 Ma.

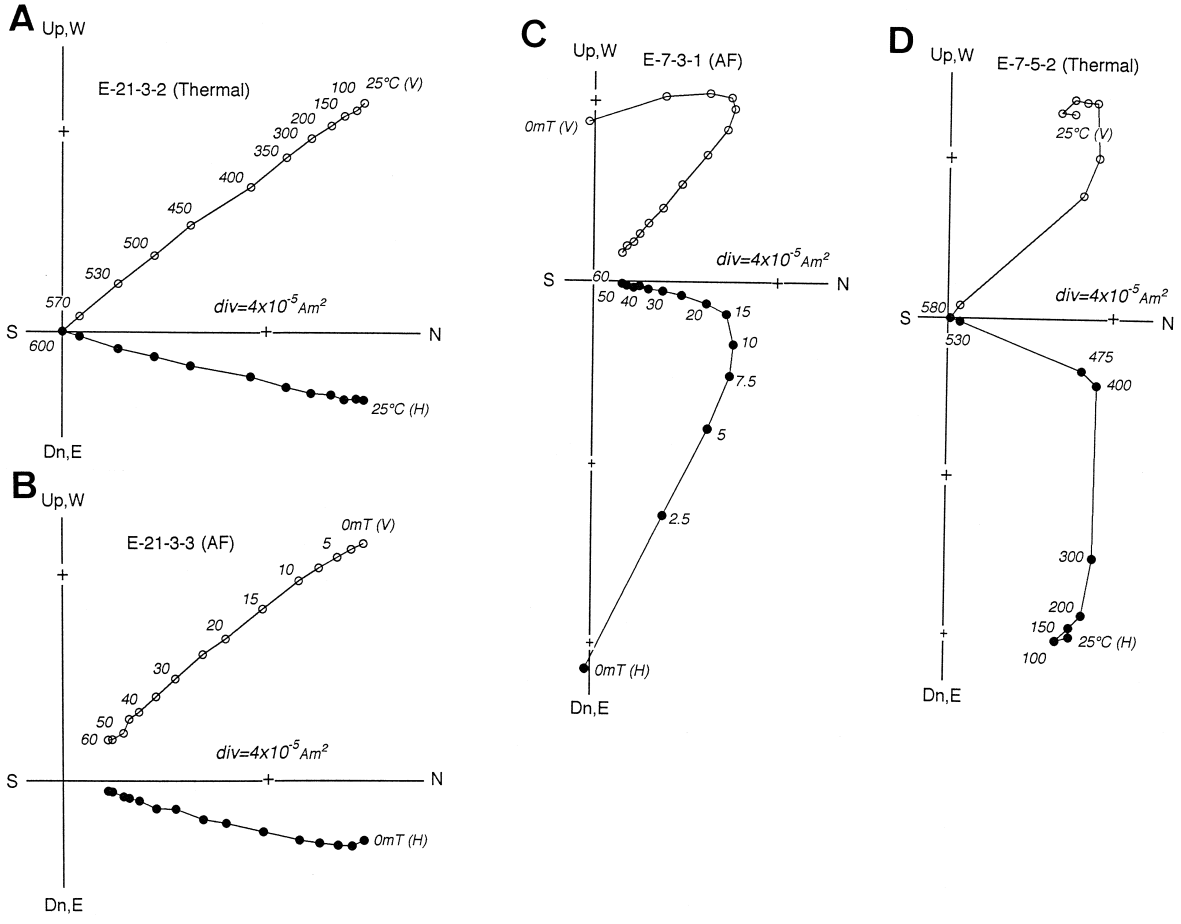


Fig. 2. Orthogonal demagnetization plots of typical specimens showing (A, B) single magnetization from site 21; (C, D) two component magnetization from site 7, revealed by thermal and alternating field demagnetization. Open (solid) circles indicate projection onto vertical (horizontal) plane.

Baker et al. (1974) and Clark and Dymond (1977) reported other older eruption ages of about 2 or 3 Ma from Poike Volcano. This suggests that Poike Volcano may have had another eruptive stage at about 3 Ma. However, the details of their age data are not clear.

Table 2
Paleomagnetic results from Easter Island

Site	N	D (°)	I (°)	k	α_{95} (°)	VGP		Sw
						Latitude, (°)	Longitude, (°)	
Poike Volcano								
E10	5	−12.7	−51.2	105	7.5	78.0	134.3	8.3
E11	5	9.8	−49.8	211	5.3	80.8	5.2	5.8
E15	4	8.6	−24.3	200	6.5	73.6	−78.5	6.0
E16	4	10.8	−19.7	873	3.1	70.2	−76.3	2.9
E17	5	6.1	−28.0	46	11.4	76.6	−83.3	12.5
E37	4	9.8	−52.7	46	13.6	79.5	19.3	12.4
E38	3	3.0	−53.4	831	4.3	82.7	50.8	3.0
E39	5	−2.1	−49.0	156	6.1	86.7	103.3	6.8
Rano Kau Volcano								
E25	5	−15.0	−41.1	387	3.9	76.0	−188.0	4.3
E26	5	−29.0	−48.8	69	9.3	64.5	147.9	10.2
E28	5	−12.0	−39.1	347	4.1	78.3	−176.3	4.5
E30	5	21.1	−43.0	40	12.2	70.9	−21.0	13.3
E33	5	−10.0	−48.7	111	7.3	80.9	142.2	8.0
E34	5	−9.8	−47.0	152	6.2	81.3	151.4	6.9
Terevaka Volcano								
E4	5	−2.3	−46.1	478	3.5	88.0	149.9	3.9
E5	3	6.9	−52.5	180	9.2	81.6	27.1	6.3
E6	4	−2.6	−43.5	328	5.1	87.1	−162.7	4.7
E7	5	2.5	−39.5	76	8.8	84.8	−83.5	9.7
E8	5	5.1	−37.4	154	6.2	82.3	−71.0	6.8
E9	5	−4.1	−47.6	355	4.1	86.0	136.5	6.9
E18	5	−2.8	−38.4	152	6.2	84.0	−134.8	6.9
E19	5	−4.0	−37.3	373	4.0	83.0	−140.5	4.4
E20	4	7.9	−43.1	450	4.3	82.6	−33.7	34.0
E21	5	13.0	−40.2	49	11.0	77.5	−36.3	12.1
E22	5	7.4	−36.1	82	8.5	80.2	−64.2	9.4
E24	4	17.4	−42.8	43	14.2	74.2	−23.7	12.9
E31	4	−25.6	−59.1	209	6.4	65.3	123.0	5.3
E32	4	−4.9	−42.1	83	10.1	84.8	−168.3	9.3
E40	5	−4.9	−49.5	66	9.5	84.6	122.2	10.4
E41	5	5.8	−40.7	110	7.3	83.6	−54.2	8.1
E42	4	0.5	−43.5	103	9.1	88.2	−93.3	8.3
E43	4	6.2	−47.9	77	10.6	84.2	1.1	9.7
E44	4	11.4	−41.1	308	5.2	79.1	−35.6	4.8
E45	4	−29.5	−51.6	146	7.6	64.0	142.2	7.0
Mean (34 sites; $k = 42.9$; $\alpha_{95} = 3.8°$)								
	4.5	0.2	−44.0	47.1	3.6	89.2	−120.9	8.0

N = number of rock samples; D = declination; I = inclination; Sw = angular dispersion at each site. The mean VGP and the Sw are calculated using individual VGP values of the 34 sites.

4. Paleomagnetism

Two or more cylindrical specimens, of 2.5 cm length and 2.5 cm diameter, were obtained from each block sample. Natural remanent magnetization (NRM) was measured using either a spinner or an ScT cryogenic magnetometer. Detailed progressive AF and thermal demagnetization were conducted on two or more pilot samples from each site; the remaining samples were AF demagnetized in at least four steps up to a maximum field of 80 mT. The principal component analysis of Kirschvink (1980) was used to calculate the best-fit demagnetization lines for linear demagnetization trajectories.

Samples from all sites have stable characteristic remanent magnetization evident from the linear decay to the origin on orthogonal plots of both AF and thermal demagnetization (Fig. 2). Unblocking temperatures of the characteristic magnetization are between 530 and 580°C. On the AF demagnetization experiments, the intensity of magnetization decreased to 30% or less after demagnetization at 70 mT. The unblocking temperature and moderate coercivities indicate that the principal magnetic carrier is magnetite or Ti-poor titanomagnetite.

In the majority of samples, except for four sites (7, 11, 24, 40), the characteristic direction is revealed after only limited demagnetization at 5 mT or 100°C. The samples from the remaining four sites have a second component of magnetization with a maximum coercivity of about 20 mT and an unblocking temperature of about 500°C. The direction of this

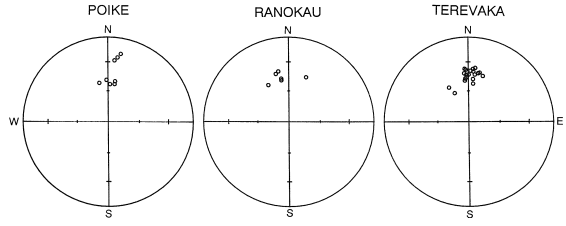


Fig. 3. Site mean paleomagnetic directions from each volcano. There is no significant difference among the directions for the three volcanoes. Equal-area projections on the upper hemisphere.

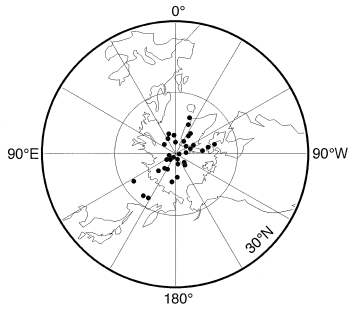


Fig. 4. Stereographic VGP plots for each site from Easter Island. The VGPs are distributes around the geographical pole.

second component differs from sample to sample and the component can be interpreted as VRM in the laboratory. We regard the characteristic component with higher coercivity and higher unblocking temperature as the primary remanence.

Reliable paleomagnetic directions with normal polarity were obtained from 34 sites (8 sites of Poike, 6 sites of Rano Kau and 20 sites of Terevaka volcanoes; Table 2, Fig. 3). The semi-angles of 95% confidence circles (α_{95}) for each site are less than 14° with a mean of 7.4°. There is no significant difference among the paleomagnetic directions of the three volcanoes. The mean direction of all 34 sites is $D = 0.2°$ and $I = -44.0°$ with an α_{95} of 3.6°. The direction is concordant with the direction obtained from Terevaka Volcano in the previous study (Isaacson and Heinrichs, 1976). The VGPs are distributed around the geographic pole (Fig. 4). The mean VGP direction is latitude = 89.2°, longitude = −120.9° and $\alpha_{95} = 3.8°$, with the α_{95} clearly containing the geographic pole.

5. Discussion

The normal polarity of the paleomagnetic directions indicates that the activity of the three volcanoes on Easter Island has been limited to the Brunhes polarity chron (0.73 Ma; Berggren et al., 1985). This supports the determined K–Ar ages which are younger than 0.7 Ma.

The angular dispersion value relative to the mean VGP was calculated from the VGPs of 34 sites after McElhinny and Merrill (1975). The 95% confidence limits of the dispersion were calculated after Eq. (11)Eq. (12) of Cox (1969). The calculated value is 11.8° with an upper limit of 14.2° and a lower limit of 9.8°.

The angular dispersion value in Easter Island is significantly lower than the values predicted by the theoretical secular variation models (Fig. 5); e.g., Model C (Cox, 1962), Model D (Cox, 1970), Model F (McFadden and McElhinny, 1984) and Model G (McFadden et al., 1988). Model G, e.g., provides a dispersion value of 14.3° at the latitude of Easter Island. This predicted value is not included in the 95% confidence limit of the observed dispersion value. The anomalously low feature of the dispersion is also indicated by the comparison between the observed value and the representative value which is calculated from a world-wide data set for the past 5 million yrs and used for the modeling calculation: the observed value of 11.8° is significantly lower than that calculated for the same latitude range (25–30°), 15.1° with lower limit of 14.4° and upper limit of 15.9° (McFadden et al., 1988).

The angular dispersion in this study is slightly lower than that in the previous study of Isaacson and Heinrichs (1976), although the difference is not significant. The larger value in the previous study is probably due to insufficient demagnetization or insufficient sampling (samples were only obtained from Terevaka Volcano). In their study, the characteristic magnetic direction in each specimen was determined by a blanket demagnetization at a single step without

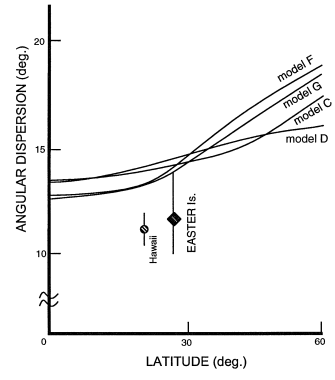


Fig. 5. Comparison of paleomagnetic angular dispersions of Easter Island and theoretical models: Model C. (Cox, 1962), Model D (Cox, 1970), Model F (McFadden and McElhinny, 1984) and Model G (McFadden et al., 1988). Angular dispersion of Hawaiian lava flows (McWilliams et al., 1982) is also indicated. The values from Easter Island and Hawaii are significantly lower than the models.

using any line fitting method. Consequently we believe that our angular dispersion value of 11.8° is a more realistic value.

The low angular dispersion of secular variation is due either to low secular variation of the geomagnetic field, to incomplete averaging of secular variation, or both. The following evidence indicates that the time span of the samples is long enough to average out the geomagnetic secular variation and to discuss the low secular variation.

First, the mean VGP of the paleomagnetic directions is in agreement with the geographic pole at the 95% confident level. The angular dispersion with respect to the geographic pole is calculated to be 11.9°. This is almost same as the value that with respect to the mean VGP (11.8°). The secular variation therefore appears to be averaged out in the mean paleomagnetic direction.

Second, the age data and the geological observation indicate that the time distribution of the samples exceeds 10^5 yrs, while the anomalously low secular variation has occurred on time scales of 10^4 yrs (Peng and King, 1992). The age data from Rano Kau Volcano are distributed over 10^5 yrs between 0.1 Ma and 0.7 Ma. From geological evidence, we estimate the time span of the samples of Terevaka and Poike Volcanoes at about 10^5 yrs, although our age data can not confirm this. It has been estimated that a time span of the order of 10^5 yrs is needed to form the total volume of Easter Island (Kaneoka and Katsui, 1985), if the eruption rate is comparable with that of the Hawaiian islands (order of tens of thousands of sq. km per million years).

We believe that the low angular dispersion is due to low levels of secular variation rather than to insufficient sampling. We conclude that the Pacific dipole window exists also at Easter Island in the southeast Pacific.

Fig. 6 shows the distribution of the area of low angular dispersions in the Pacific with the Pacific dipole window extending over the whole of the Pacific ocean. Although the data from Society Is-

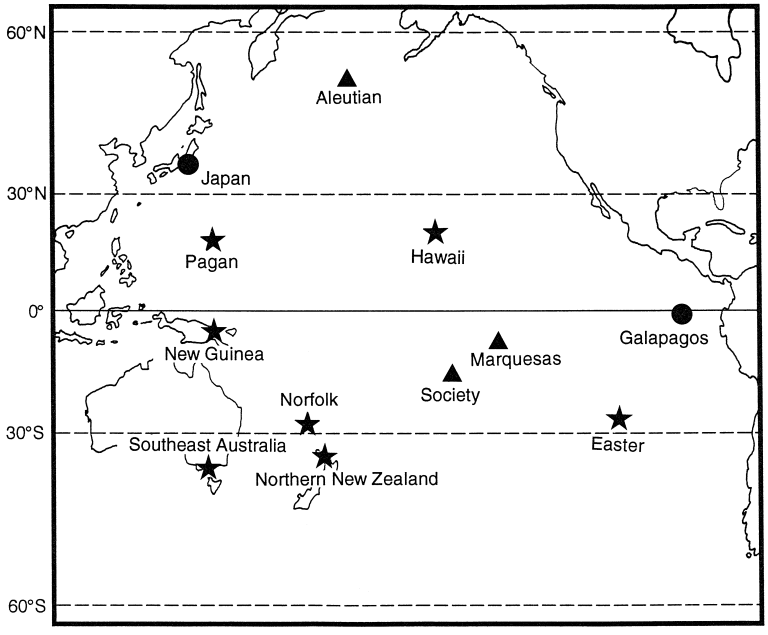


Fig. 6. Distribution of the low angular dispersion area on the Pacific region. Stars indicate the area where low angular dispersions are observed; Pagan (U.S.–Japan Paleomagnetic Cooperation Program in Micronesia, 1975), Hawaii (e.g., McWilliams et al., 1982), New Guinea (Manwaring, 1971), Norfolk (Aziz-ur-Rahman and McDougall, 1973), Southeast Australia (Aziz-ur-Rahman, 1971), Northern New Zealand (Shibuya et al., 1995), and Easter Island (this study). Angular dispersions are concordant with the model's value in areas with circles; Japan (Heki, 1983) and Galapagos (Doell and Cox, 1972). Triangles indicate the areas where the angular dispersions recalculated by Shibuya et al. (1995) are lower than the models.

lands (Duncan, 1975), Hiva-Oa Island (Katao et al., 1988) and Galapagos (Doell and Cox, 1972) do not indicate the existence of Pacific dipole window, the recalculated angular dispersions using the composite Fisher method indicate the existence of the Pacific dipole window in these areas (Shibuya et al., 1995).

The Pacific dipole window appears to be the local expression of a global rather than simply a local phenomenon. This long-wavelength anomaly of the secular variation can arise from a global-scale lateral inhomogeneity at the lower mantle or the core–mantle boundary as discussed in the recent studies; i.e., existence of high electrical conductivity layer under the Pacific (Runcorn, 1992), and coupling of the core fluid flow to the mantle (e.g., Bloxham and Gubbins, 1985, 1987).

Acknowledgements

We thank Y. Otake and K. Takatsugi who kindly helped us in the laboratory work. Dr. Simon Wallis provided useful advice. We also thank anonymous referees for helpful reviews of this manuscript. This work was supported by the Ministry of Education, Science and Culture Grant-in-Aid for Overseas Scientific Survey No. 58041049 and 59043044.

References

Aziz-ur-Rahman, 1971. Palaeomagnetic secular variation for recent normal and reversed epochs, from the newer volcanics of Victoria, Australia. Geophys. J. R. Astr. Soc. 24, 255–269.

Aziz-ur-Rahman, McDougall, I., 1973. Palaeomagnetism and palaeosecular variation on lavas from Norfolk and Philip Islands, South-west Pacific Ocean. Geophys. J. R. Astr. Soc. 33, 141–155.

Baker, P.E., 1966. Preliminary account of recent geological investigations on Easter Island. Geol. Mag. 104, 116–122.

Baker, P.E., Buckley, F., Holland, J.G., 1974. Petrology and geochemistry of Easter Island. Contr. Mineral. Petrol. 44, 85–100.

Berggren, W.A., Kent, D.V., Flynn, J.J., Couvering, J.A.V., 1985. Cenozoic geochronology. Geol. Soc. Am. Bull. 96, 1407–1418.

Bloxham, J., Gubbins, D., 1985. The secular variation of Earth's magnetic field. Nature 317, 777–781.

Bloxham, J., Gubbins, D., 1987. Thermal core–mantle interactions. Nature 325, 511–513.

Clark, J.G., Dymond, J., 1974. Age, chemistry, and tectonic significance of Easter Island. EOS Trans. Am. Geophys. Union 55, 300.

Clark, J.G., Dymond, J., 1977. Geochronology and petrochemistry of Easter and Sala y Gomez Islands: implication for the origin of the Sala y Gomez Ridge. J. Volcanol. Geotherm. Res. 2, 29–48.

Cox, A., 1962. Analysis of present geomagnetic field for comparison with paleomagnetic results. J. Geomag. Geoelectr. 13, 101–112.

Cox, A., 1969. Confidence limits for the precision parameter κ. Geophys. J. R. Astr. Soc. 18, 545–549.

Cox, A., 1970. Latitude dependence of the angular dispersion of the geomagnetic field. Geophys. J. R. Astr. Soc. 20, 253–269.

Doell, R.R., Cox, A., 1961. Paleomagnetism of Hawaiian lava flows. Nature 192, 645–646.

Doell, R.R., Cox, A., 1963. The accuracy of the paleomagnetic method as evaluated from historic Hawaiian lava flows. J. Geophys. Res. 68, 1997–2009.

Doell, R.R., Cox, A., 1965. Paleomagnetism of Hawaiian lava flows. J. Geophys. Res. 70, 3377–3405.

Doell, R.R., Cox, A., 1971. Pacific geomagnetic secular variation. Science 171, 248–254.

Doell, R.R., Cox, A., 1972. The Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle. In: Robertson, E.C. (Ed.), Nature of the Solid Earth. McGraw-Hill, New York, 245 pp.

Duncan, R.A., 1975. Palaeosecular variation at the Society Islands, French Polynesia. Geophys. J. R. Astr. Soc. 41, 245–254.

Gonzalez-Ferran, O., Cordani, G.U., Halpern, M., 1974. Potassium-argon ages and Sr^{37}/Sr^{36} ratios of volcanic rocks from Easter Island. Abstracts of papers, IAVCEI International Symp. on Volcanology. Andean and Antarctic Volcanology Problems, 28 pp.

Heki, K., 1983. Paleomagnetic study of the Higashi–Izu monogenetic volcano group and pyroclastic flow deposits in Kagoshima Prefecture: paleosecular variation during the last 40,000 years in Japan. J. Geomag. Geoelectr. 35, 383–390.

Isaacson, L.B., Heinrichs, D.F., 1976. Paleomagnetism and secular variation of Easter Island basalt. J. Geophys. Res. 81, 1476–1482.

Kaneoka, I., Katsui, Y., 1985. K–Ar ages of volcanic rocks from Easter Island. Bull. Volcanol. Soc. Jpn. Ser. 2 30, 33–36.

Katao, H., Morinaga, H., Hyodo, M., Inokuchi, H., Matsuda, J., Yaskawa, K., 1988. Geomagnetic paleosecular variation and K–Ar ages in Hiva-Oa Island, Marquesas, French Polynesia. J. Geomag. Geoelectr. 40, 703–714.

Kirschvink, J.L., 1980. The least-square line and plane, and the analysis of palaeomagnetic data. Geophys. J. R. Astr. Soc. 62, 699–718.

Mankinen, E.Q., Darlymple, G.B., 1972. Electron microscope evaluation of terrestrial basalts for whole-rock K–Ar dating. Earth Planet. Sci. Lett. 17, 89–94.

Manwaring, E.A., 1971. Paleomagnetism of some recent basalts from New Guinea. Aust. Bur. Miner. Resour. Geol. Geophys. 45, 23.

McElhinny, M.W., Merrill, R.T., 1975. Geomagnetic secular variation over the past 5 m.y. Rev. Geophys. Space Phys. 13, 687–708.

McFadden, P.L., McElhinny, M.W., 1984. A physical model for paleosecular variation. Geophys. J. R. Astr. Soc. 78, 809–830.

McFadden, P.L., Merrill, R.T., McElhinny, M.W., 1988. Dipole/quadrupole family modeling of paleosecular variation. J. Geophys. Res. 93, 11583–11588.

McWilliams, M.O., Holcomb, R.T., Champion, D.E., 1982. Geomagnetic secular variation from ^{14}C-dated lava flows on Hawaii and the question of the Pacific non-dipole low. Phil. Trans. R. Soc. London A 306, 211–222.

Nagao, K., Itaya, T., 1988. K–Ar age determination. Mem. Geol. Soc. Jpn. 29, 5–21, in Japanese with English abstract.

Peng, L., King, J.W., 1992. A late Quaternary geomagnetic secular variation record from Lake Waiau, Hawaii, and the question of the Pacific nondipole low. J. Geophys. Res. 97, 4407–4424.

Runcorn, S.K., 1992. Polar path in geomagnetic reversals. Nature 356, 654–656.

Shibuya, H., Cassidy, J., Smith, I.E.M., Itaya, T., 1995. Paleomagnetism of young New Zealand basalts and longitudinal distribution of paleosecular variation. J. Geomag. Geoelectr. 47, 1011–1022.

Steiger, R.H., Jäger, E., 1977. Subcommission on geochronology: Convention on the use of decay constants in geo-and cosmochronology. Earth Planet. Sci. Lett. 36, 359–362.

Terashima, S., 1970. Determination of sodium, potassium magnesium, calcium, manganese and iron in the silicate rocks and terrestrial water by atomic absorption spectrophotometry. Bull. Geol. Surv. Jpn. 21, 693–707, (in Japanese with English abstract).

U.S.–Japan Paleomagnetic Cooperation Program in Micronesia, 1975. Paleosecular variation of lavas from the Marianas in the western Pacific Ocean. J. Geomag. Geoelectr. 27, 57–66.